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Lab #5 part 1
4201 L

Lab 5 part 1: A to D converter LabVIEW

Introduction

In this lab we built a LabVIEW VI to model an ideal ADC and display the stair steps. Using labVIEW's software, I can manipulate the FSR and number of bits to change the Q value

$(Q = \frac{FSR}{2^m})$. The maximum error in A-D is half a step, and can be calculated as $\frac{Q}{2} = \frac{FSR}{2^{m+1}}$.

Essentially I built a function where the input(user determined) is the number of bits and desired FSR, plus the analog input voltage. Q is the most important factor and is essentially our resolution and digitizing factor as it is the step size, or how much input voltage each digital step represents. By changing the Q value we are impacting the rate, range and resolution. Using the created LabVIEW program we can see how manipulating the Q in various ways impacts the overall digital output.

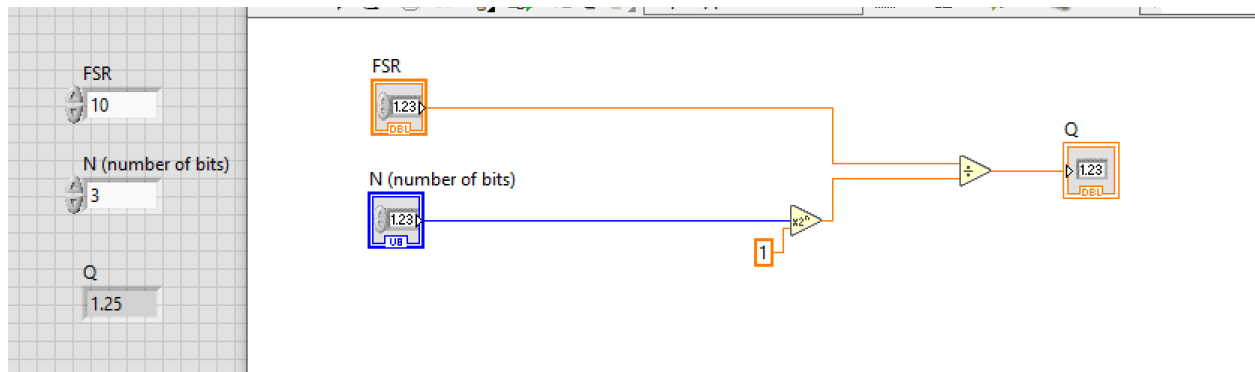
1 & 2. First determine $Q = FSR/2^n$ where n is the number of bits in the system and $Q/2$ will be calculated and used in the equation: Suggestion use the 3 bit ADC and FSR = 10V example from the handout.

Digital # or digital counts = (input voltage + $Q/2$)/Q round down the result to the nearest integer, no half bit in digital systems

$Q = \frac{FSR}{2^m}$, where

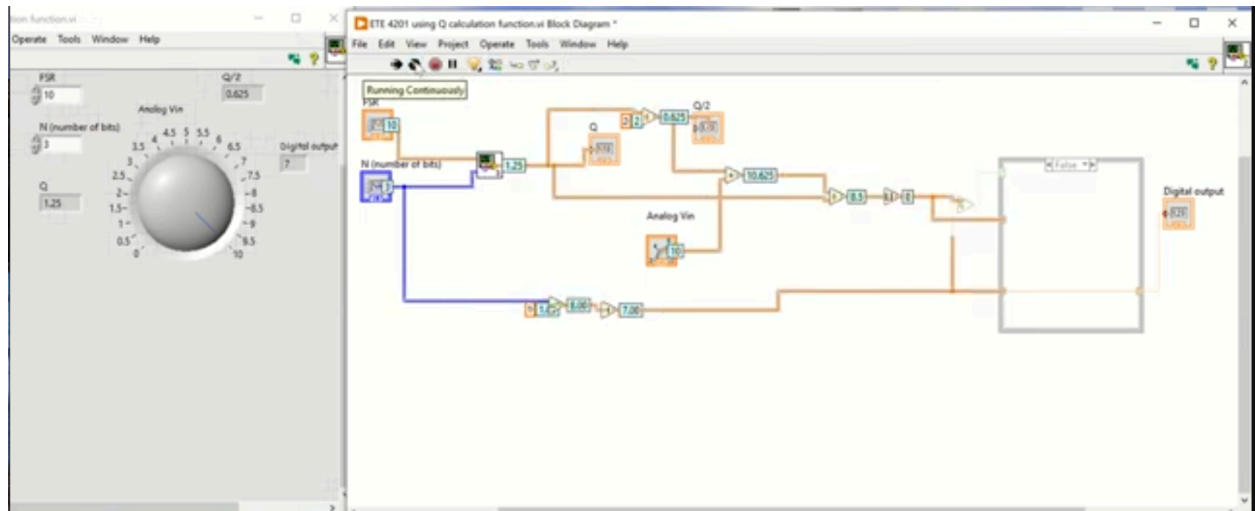
- Q is the step size, or how much input voltage each digital step represents.
- FSR is the full scale range, or the total input voltage the ADC can handle. If the analog to digital converter reads 0 to 10v, then FSR= 10v.
- N is the number of bits, or how many bits the digital output has. For three bits, the output codes go to 2^3 or from 0 to 7.

$$Q = \frac{FSR}{2^m} = \frac{10V}{2^3} = 1.25$$



IMG 1. Calculations using LabView. The calculated Q value is 1.25.

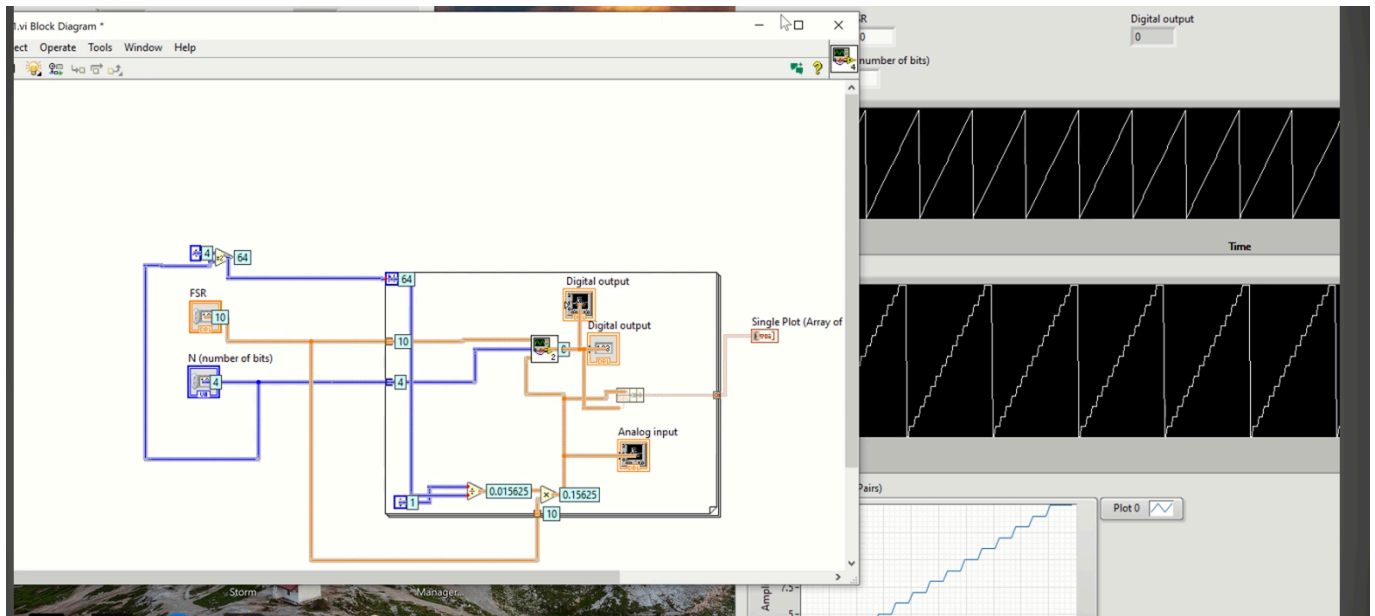
3. Create a VI that plots the A/D function across the full range of input voltages. This vi will sweep the analog input voltage from 0 to FSR calculating the digital count and then plotting the result. For the number of points to plot use 4 times the number of A/D steps: i.e. an 8 bit ADC has 256 steps (0-255) so 4 x (256) or 1024 points will provide a good data set to plot. For plotting hints check out the XY Graph examples in the help files.



IMG 2. Calculating digital output portion of the LabVIEW program

Link: <https://www.youtube.com/watch?v=fqEIS5mgwms>

4. Plot the error difference between the actual input value and the value represented by the digital code of the ADC. Confirm that error ranges between plus and minus $Q/2$ but near FSR range the ADC error becomes $3Q/2$ or $1.5Q$.



IMG 3. Full LabVIEW program, viewable at the link below.

Link: https://www.youtube.com/watch?v=_elQWpqD2wY